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BATCH - 2023

Q1 - #include <iostream>

using namespace std;

int main() {

 cout << "Hello World";

 return 0;

}

A screenshot of a terminal window with a dark background. The terminal shows the following commands and output:

```
cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION % cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
Hello World
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %
```

 The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active. On the right side of the terminal window, there are icons for Code, a plus sign, a square, a trash can, and a close button.

Q2 - #include <iostream>

using namespace std;

int main() {

 int a, b;

 cin >> a >> b;

 if (a > b)

 cout << a;

 else

 cout << b;

 return 0;

}

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION % cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Des
1
2
23
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %
```

Q3 - #include <iostream>

using namespace std;

int main() {

float a, b, c;

cin >> a >> b >> c;

float x = a / b - c;

cout << x;

return 0;

}

```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION % cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Des
1
23
3
-2.95652
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %
```

Q4 - #include <iostream>

using namespace std;

int main() {

cout << "Hello\nWorld";

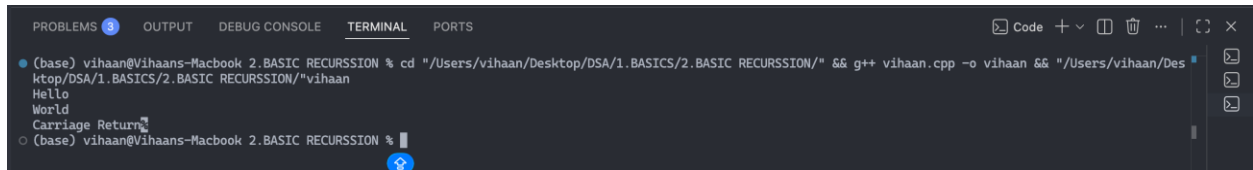
cout << endl << "Hello\tWorld";

cout << "\a";

cout << "\rCarriage Return";

return 0;

```
}
```



```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION % cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Des
ktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
Hello
World
Carriage Return
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %
```

Q5 - #include <iostream>

using namespace std;

namespace First {

int value = 10;

void show() {

cout << value << endl;

}

}

namespace Second {

int value = 20;

void show() {

cout << value << endl;

}

}

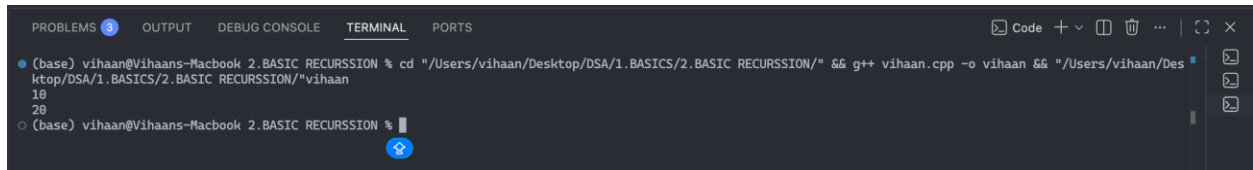
int main() {

First::show();

Second::show();

return 0;

```
}
```



The screenshot shows a terminal window with the following text: (base) vihaan@Vihaans-Macbook 2.BASIC RECURSION % cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan 10 20. The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active. There are icons for Code, a plus sign, a square, a trash can, and a refresh button in the top right corner of the terminal window.

Q6 - #include <iostream>

#include <string>

using namespace std;

struct Student {

 string name;

 int roll;

 float marks;

void setStudentData() {

 cin >> name >> roll >> marks;

}

void getStudentData() {

 cout << name << " " << roll << " " << marks << endl;

}

};

int main() {

 Student s[3];

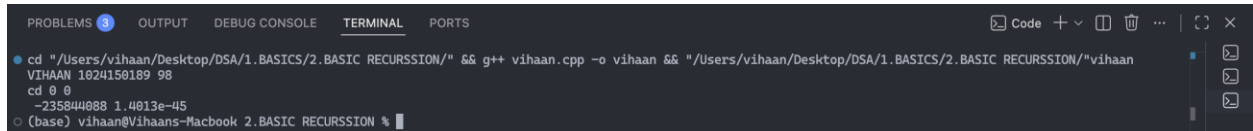
```

    for (int i = 0; i < 3; i++) {
        s[i].setStudentData();
    }

    for (int i = 0; i < 3; i++) {
        s[i].getStudentData();
    }

    return 0;
}

```



```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
VIHAAN 1024150189 98
cd 0 0
-235844088 1.4013e-45
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %

```

Q7 - #include <iostream>

#include <string>

using namespace std;

class Student {

private:

string name;

int roll;

float marks;

public:

void setStudentData() {

cin >> name >> roll >> marks;

}

```

void getStudentData() {
    cout << name << " " << roll << " " << marks << endl;
}
};

```

```

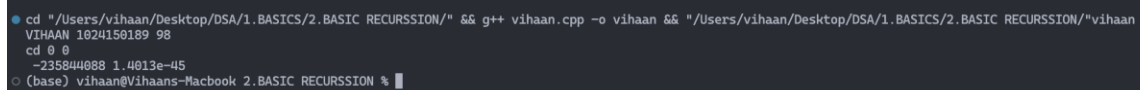
int main() {
    Student s[3];

    for (int i = 0; i < 3; i++) {
        s[i].setStudentData();
    }

    for (int i = 0; i < 3; i++) {
        s[i].getStudentData();
    }

    return 0;
}

```



```

● cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
VIHAAN 1024150189 98
cd 0 0
-235844088 1.4013e-45
○ (base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %

```

Q8 - #include <iostream>

#include <string>

using namespace std;

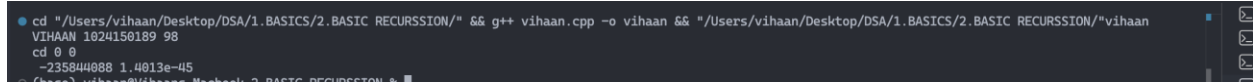
```
class Student {  
    string name;  
    int roll;  
    float marks;  
  
public:  
    void setStudentData() {  
        cin >> name >> roll >> marks;  
    }  
  
    void getStudentData() {  
        cout << name << " " << roll << " " << marks << endl;  
    }  
};  
  
int main() {  
    Student s[3];  
  
    for (int i = 0; i < 3; i++) {  
        s[i].setStudentData();  
    }  
  
    for (int i = 0; i < 3; i++) {  
        s[i].getStudentData();  
    }
```

```

return 0;

}

```



```

cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
VIHAAN 1024150189 98
cd 0 0
-235844088 1.4013e-45

```

Q9 - In C, structures are pretty simple — they're mainly just a way to bundle related variables together, and that's it. You can't put functions inside them, you don't get any access control, and everything is public by default. In C++, structures are much more powerful and feel a lot like classes. You can have functions inside a structure, control access using public and private, and even use object-oriented concepts like encapsulation and inheritance. Basically, C structs are just data holders, while C++ structs can actually behave like real objects.

Q10 - #include <iostream>

using namespace std;

```

int main() {

    int n;

    cin >> n;

    for (int i = 0; i < n; i++) {

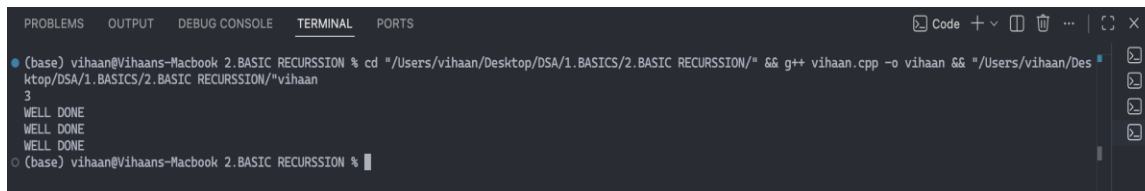
        cout << "WELL DONE" << endl;

    }

    return 0;

}

```



```

(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION % cd "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/" && g++ vihaan.cpp -o vihaan && "/Users/vihaan/Desktop/DSA/1.BASICS/2.BASIC RECURSION/"vihaan
3
WELL DONE
WELL DONE
WELL DONE
(base) vihaan@Vihaans-Macbook 2.BASIC RECURSION %

```